

AI RESEARCH ASSISTANT AGENT

AN AGENTIC AI-POWERED RESEARCH & LEARNING
PLATFORM

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Designed for : **AI RESEARCH ASSISTANT AGENT**

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PROBLEM

- Learning a new tech topic means hunting across many scattered sources
- Manually building explanations, quizzes & interview prep takes hours
- Generic AI chatbots give one-off answers, not a structured study path
- No single place to track research history and revision progress

SOLUTION

- Autonomous AI research agent
- 8-step agentic workflow per topic
- Interactive MCQ quiz generator
- Interview prep & learning roadmap
- One-click PDF report export

UNIQUE VALUE PROPOSITION

- Turns any topic into a complete study package — explanation, key points, interview Q&A, quiz, roadmap and references — not a single chatbot reply.

UNFAIR ADVANTAGE

- Full 8-step chained agent pipeline, not a single prompt
- Real-time SSE progress streaming
- Context-aware prompting across steps
- Automated dual-pass ReportLab PDF export

CUSTOMER SEGMENT

- Students preparing for exams
- Job seekers prepping for interviews
- Self-learners exploring new topics
- Educators & coaching institutes

COST STRUCTURE

- Gemini API usage
- Cloud hosting
- Database services
- Domain & SSL
- Development & maintenance

REVENUE STREAMS

- Freemium / premium subscription
- Institutional licensing (colleges, coaching classes)
- API access for partner platforms
- Sponsored reference & partner links

INTRODUCTION

Understanding a new technical subject usually means jumping between blogs, videos, and outdated notes just to piece together a coherent study plan. The AI Research Assistant Agent is an Agentic AI-powered web application that takes a single topic typed in by the user and autonomously builds a complete, structured learning package around it. Rather than acting like a simple chatbot that answers one question at a time, it runs an eight-step research pipeline, streaming its progress live, and produces an explanation, key points, interview questions, a quiz, a summary, a learning roadmap and references — all in one pass.

PROBLEM STATEMENT

- Learning a new technical subject requires piecing information together from many scattered sources.
- Manually writing explanations, quizzes and interview questions for self-study is slow and tedious.
- Generic AI chatbots answer one question at a time with no structured learning journey.
- Users struggle to know what to study next or how to prepare systematically for interviews.
- There is no single tool that researches, organizes, quizzes and documents a topic end-to-end.

OBJECTIVE

- Accept any research topic and autonomously plan a full study pipeline.
- Generate a clear, structured technical explanation using Gemini.
- Extract high-impact key points and takeaways automatically.
- Produce interview-style questions relevant to the topic.
- Generate an interactive multiple-choice quiz with instant feedback.
- Summarize the entire research session into an executive summary.
- Build a step-by-step learning roadmap and suggest useful references.
- Let users export the full report as a professional PDF and revisit past searches.

KEY FEATURES

- Secure user authentication (login / register / logout)
- 8-step autonomous research agent pipeline
- Live progress via Server-Sent Events (SSE)
- In-depth AI-generated technical explanations
- Auto-extracted key points & takeaways
- Interview question generator
- Interactive MCQ quiz with instant feedback
- Learning roadmap & curated references
- ChatGPT-style contextual chat follow-ups
- One-click PDF export (ReportLab)
- Dashboard with search history & analytics charts
- Dark / light glassmorphism theme switch

THE 8-STEP AGENTIC WORKFLOW

The agent chains each step's output into the next step's prompt.

1. Understand the user's query

2. Generate a detailed explanation

3. Generate key points

4. Generate interview questions

5. Generate MCQ quiz with answers

6. Generate an executive summary

7. Generate a learning roadmap

8. Suggest useful references

TECHNOLOGY STACK

Frontend	HTML5, CSS3 (custom variables & themes), Vanilla JavaScript, Server-Sent Events, Chart.js
Backend	Python Flask, Blueprint-based MVC routing
Database	SQLite (users, searches, reports, timestamps)
AI Engine	Google Gemini API with dedicated prompts for explanation, quiz, interview prep, roadmap and summary
PDF Engine	ReportLab with a custom dual-pass NumberedCanvas for accurate page counts
Design	Glassmorphism UI with full dark / light theme support, responsive layout

ARCHITECTURE — CLEAN MVC PATTERN

VIEWS

templates/ — base, landing, dashboard, agent workspace, auth pages (Jinja2)

CONTROLLERS

routes/ — auth.py, dashboard.py, agent.py (Flask blueprints)

SERVICES

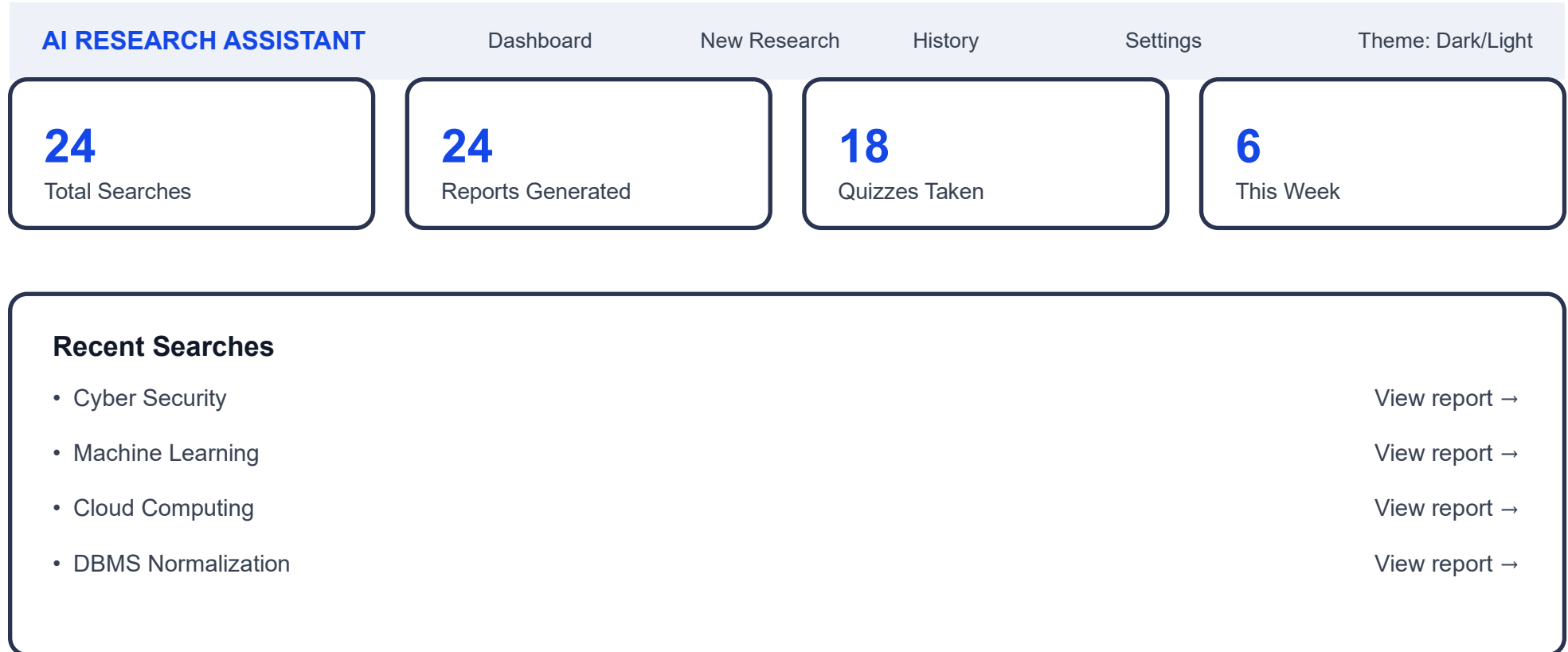
services/ — ai_service.py (Gemini prompts), pdf_service.py (ReportLab export)

MODEL / DB

database.py — SQLite schema for users, searches, reports

WORKSPACE PREVIEW — DASHBOARD (WIREFRAME)

Illustrative layout — replace with live screenshots once the app is running.



WORKSPACE PREVIEW — RESEARCH AGENT (WIREFRAME)

Agent Progress

- Understand query
- Explanation
- Key points
- Interview prep
- MCQ quiz
- Summary
- Roadmap
- References

Chat with your report

Explain public key vs private key in simple terms.

Public key encryption uses two mathematically linked keys: a public key anyone can use to encrypt data, and a private key only the owner holds to decrypt it...

Give me 2 interview questions on this.

GENERATIVE AI vs. AGENTIC AI

Generative AI

A model takes one prompt and produces one piece of content in response — for example, asking Gemini to explain a term or write a set of MCQs.

Agentic AI

The system plans and executes a multi-step workflow on its own: it chains eight tasks together, feeding each output into the next prompt, streams live progress, and can answer contextual follow-up questions about the finished report.

CONCLUSION

The AI Research Assistant Agent shows how Agentic AI can move beyond a simple question-answer chatbot into a genuine autonomous research assistant. By chaining eight coordinated steps, streaming live progress, and packaging the results into explanations, quizzes, interview prep, roadmaps and a downloadable PDF, the platform turns any topic into a complete, structured study resource — while still encouraging users to verify important facts independently.

THANK YOU

AI RESEARCH ASSISTANT AGENT

"Ask once. Get the whole study plan."

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